

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458512.61955 +/- 0.00049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.043
Transit depth (Rp/Rs)^2	2.3 +/- 1.3 [%]
Orbital Inclination (inc)	83.73 +/- 2.6
Airmass coefficient 1 (a1)	0.687 +/- 0.019
Airmass coefficient 2 (a2)	0.26 +/- 0.18
Scatter in the residuals of the lightcurve fit is	2.69 %
Best Comparison Star	#3 - [366, 375]
Optimal Aperture	3.08
Optimal Annulus	7.55
Transit Duration (day)	0.046 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.152 $\pm$ 0.043 vs 0.11 (deviation 0.98 σ)
Duration fit vs published	0.046 d vs 0.1208 d (deviation 2.08 σ)
Mid-time O-C	-0.8 min
Depth SNR	3.5
Residual scatter	2.69 %

Light curve

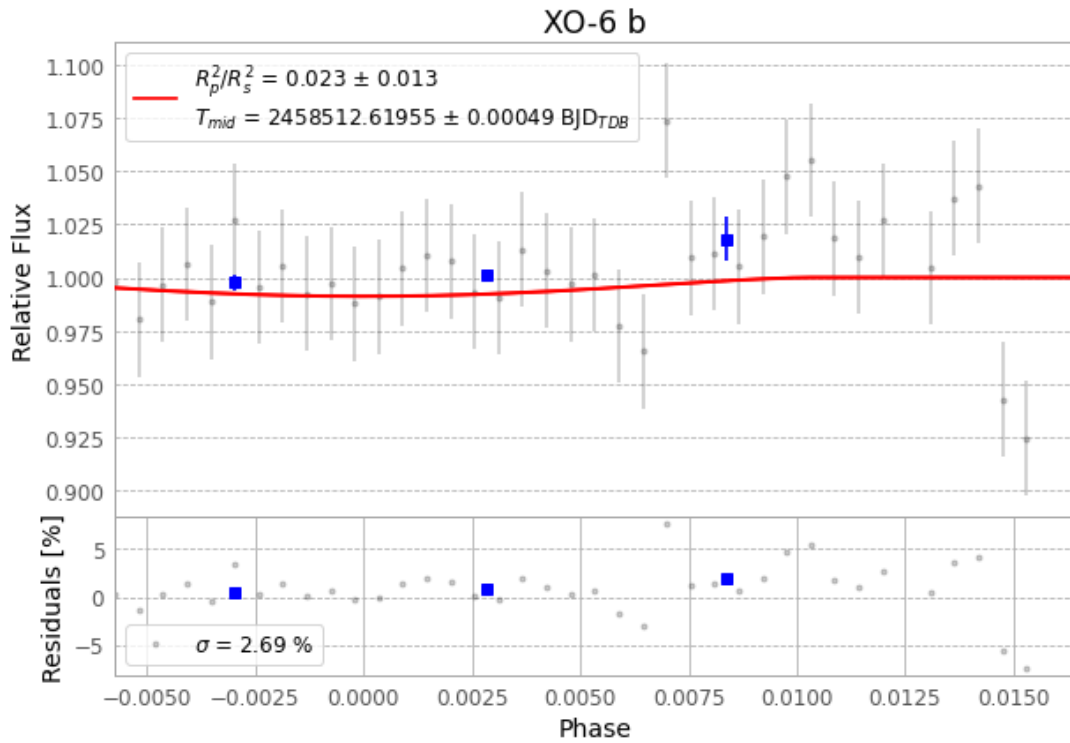


Figure 1 — FinalLightCurve_XO-6 b_2019-01-28.png (SHA-256 7c748010e9b9dcda...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [564, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a98f58db339c2bdd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

44 FITS frames (29 MB), XO-6190129021512.FITS → XO-6190129042412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-190129.log (c15c47bc2779...), FinalLightCurve_XO-6 b_2019-01-28.png (7c748010e9b9...), FinalLightCurve_XO-6 b_2019-01-28.csv (505fda5d052e...)

Compute resources

Wall time 128.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:00Z.